Online Estimation of Average Gradient Outer Product Alignment During Grokking:

Formal Theory, Perturbation Analysis, and Self-Calibrated Extrapolation

Abstract

Evaluating progress toward delayed generalization (grokking) via the Frobenius cosine similarity between the current Average Gradient Outer Product (AGOP) G_t and the asymptotic grokked AGOP G_T requires offline access to the end-of-training operator G_T. In this document, we formulate a testable reduced-order geometric model of post-interpolation AGOP dynamics on the spectraplex and derive a closed-form online alignment estimator $\widehat{s}_{\mathsf{spiked}}(t)$ driven by the fractional compression in effective rank relative to the post-interpolation reference operator G_0. We establish exact non-asymptotic angle-addition laws linking $\widehat{s}_{\mathsf{spiked}}(t)$ and the baseline-projection estimator $\widehat{s}_{\mathsf{proj}}(t)$ to the offline alignment $s(t) = \mathsf{cos}(G_t, G_T)$, characterize the orthogonality defect δ and its associated purity dead zone when $\delta < 0$, introduce online diagnostics that separately test trajectory collinearity ($D_{\mathsf{chord}}(t)$), baseline orthogonality drift ($v(t)$), and exponential rate stationarity ($D_{\mathsf{exp}}(t)$), and prove a window-averaged three-horizon exponential extrapolation theorem that recovers the asymptotic effective rank r, the calibrated alignment $s(t)$, and the predicted alignment-threshold time t^* online for both $\delta = 0$ and $\delta \neq 0$.

1 Preliminaries, Operator Definitions, and Causal Anchoring

Geometric Intuition and Motivation. The fundamental obstacle to using $s(t) = \mathsf{cos}(G_t, G_T)$ as an online early-stopping or grokking-detection metric is that G_T lies in the causal future of time t. However, during grokking, the network transitions from a high-rank memorizing regime at interpolation to a low-rank feature-learning regime as $t \to \infty$. By projecting G_t onto the trace-normalized spectraplex $\Delta^{d\times d}$, we decouple overall gradient-norm scaling from structural feature learning. On $\Delta^{d\times d}$, the trace is fixed to unity ($\mathsf{Tr}(\tilde{G}_t) = 1$), so the squared Frobenius norm $\|\tilde{G}_t\|_F^2 = \mathsf{Tr}(\tilde{G}_t^2)$ measures purely the concentration of the eigenvalue spectrum (spectral purity $\mathcal{P}(t)$). If $\tilde{G}_t$ moves from a high-rank reference state $\tilde{G}_0$ toward a low-rank state $\tilde{G}_T$ along a direction roughly orthogonal to $\tilde{G}_0$, the Pythagorean theorem links the unobservable forward angle $\mathsf{cos}(G_t, G_T)$ directly to the observable backward norm ratio $\|\tilde{G}_0\|_F/\|\tilde{G}_t\|_F = \sqrt{\mathcal{P}_0/\mathcal{P}(t)}$.

Let $\mathcal{X} \subset \mathbb{R}^d$ denote the input space and let $f_t : \mathbb{R}^d \to \mathbb{R}^c$ be a differentiable neural network parameterized by weights θ_t at training time $t \geq 0$. Given a training dataset $\{x_i\}_{i=1}^N \subset \mathcal{X}$, let $J_t(x_i) = \nabla_x f_t(x_i) \in \mathbb{R}^{c\times d}$ denote the input-output Jacobian evaluated at sample x_i.

Definition 1.1 (Average Gradient Outer Product and Spectraplex Projection)**.** Let $\mathcal{S}^{d\times d}$ denote the space of $d \times d$ real symmetric matrices and $\mathcal{S}_+^{d\times d} \subset \mathcal{S}^{d\times d}$ the cone of positive semidefinite (PSD) matrices, equipped with the Frobenius inner product $\langle A, B\rangle_F := \mathsf{Tr}(A^\top B)$ and norm $\|A\|_F := \sqrt{\mathsf{Tr}(A^2)}$. The Average Gradient Outer Product (AGOP) at time t is defined as:

$$G_t := \frac{1}{N}\sum_{i=1}^{N} J_t(x_i)^\top J_t(x_i) \in \mathcal{S}_+^{d\times d} \setminus \{0\}. \tag{1}$$

We define the trace-normalized AGOP $\tilde{G}_t$ via projection onto the spectraplex $\Delta^{d\times d} := \{X \in \mathcal{S}_+^{d\times d} : \mathsf{Tr}(X) = 1\}$:

$$\tilde{G}_t := \frac{G_t}{\mathsf{Tr}(G_t)}, \qquad \mathsf{Tr}(\tilde{G}_t) = 1. \tag{2}$$

Definition 1.2 (Causal Interpolation Anchor and Asymptotic Grokked Operator)**.** Because regularized training objectives (such as cross-entropy loss under positive weight decay) do not attain exact zero loss in finite time, the interpolation time t_{mem} is defined causally as the first hitting time of a prescribed training-loss threshold $\epsilon_{\mathsf{train}} > 0$ (or saturation of training accuracy at 1):

$$t_{\mathsf{mem}} := \inf\left\{t \geq 0 : \mathcal{L}_{\mathsf{train}}(t) \leq \epsilon_{\mathsf{train}}\right\}. \tag{3}$$

To suppress mini-batch stochasticity at interpolation and establish a single consistent starting anchor, we average G_τ over a short causal window $[t_{\mathsf{mem}}, t_0]$ of width $w > 0$, where $t_0 := t_{\mathsf{mem}} + w$ denotes the **anchor time** at which post-interpolation tracking begins:

$$G_0 := \frac{1}{w}\int_{t_{\mathsf{mem}}}^{t_0} G_\tau \, d\tau, \qquad \tilde{G}_0 := \frac{G_0}{\mathsf{Tr}(G_0)} \in \Delta^{d\times d}. \tag{4}$$

Whenever $w \ll 1/\kappa$ (where κ is the post-interpolation relaxation rate in Section 5), the AGOP is quasi-stationary inside $[t_{\mathsf{mem}}, t_0]$ so that $\tilde{G}_{t_0} \approx \tilde{G}_0$. Because κ is estimated only after t_0, in practice one fixes a short window width w a priori and verifies a posteriori that the in-window drift offset is negligible via $\widehat{c}(t) \approx 1$ and $D_\alpha(t) \approx 0$ in Theorem 5.2; if a non-zero offset $\alpha_{t_0} = \alpha_0 \in [0, 1)$ is present, it is accommodated automatically by the three-window Aitken extrapolation in Lemma 5.1. We define the **asymptotic grokked operator** $\tilde{G}_T \in \Delta^{d\times d}$ as the long-time limit $\tilde{G}_T := \lim_{t\to\infty} \tilde{G}_t$ (with G_T denoting any representative positive scalar multiple along the terminal ray). At any finite stopping horizon $T \in (t_0, \infty)$, $\tilde{G}_t$ is close to $\tilde{G}_T$ up to an exponentially small residual controlled by the relaxation rate in Section 5.

Definition 1.3 (Offline AGOP Alignment)**.** By the invariance of Frobenius cosine similarity to positive scalar multiplication, the offline AGOP alignment metric $s(t)$ for $t \geq t_0$ is given equivalently in terms of G_t or $\tilde{G}_t$:

$$s(t) := \cos(G_t, G_T) = \frac{\langle G_t, G_T\rangle_F}{\|G_t\|_F\|G_T\|_F} = \frac{\langle \tilde{G}_t, \tilde{G}_T\rangle_F}{\|\tilde{G}_t\|_F\|\tilde{G}_T\|_F}. \tag{5}$$

Remark 1.1 (Distinction from Rooted Feature Matrix Alignment)*.* In the Neural Feature Ansatz (NFA) literature (e.g., Mallinar et al.), feature alignment is evaluated on the matrix square root $M_t := G_t^{1/2}$ via $s_{\mathsf{NFA}}(t) := \cos(G_t^{1/2}, G_T^{1/2})$. Because the operator square root $X \mapsto X^{1/2}$ is nonlinear on $\mathcal{S}_+^{d\times d}$, a linear chord $(1-\alpha_t)\tilde{G}_0 + \alpha_t\tilde{G}_T$ in $\tilde{G}_t$ does not induce a linear chord in $G_t^{1/2}$ unless $G_0G_T = 0$; even when $G_0G_T = 0$, taking the square root maps the mixing weights as $(\alpha_t, 1-\alpha_t) \mapsto (\sqrt{\alpha_t}, \sqrt{1-\alpha_t})$ prior to trace normalization, yielding a chord with a nonlinearly reparameterized time coordinate. Furthermore, $\mathsf{Tr}(G_t^{1/2})$ depends on the $\ell_{1/2}$ quasi-norm of the spectrum of G_t, whereas $\|G_t^{1/2}\|_F^2 = \mathsf{Tr}(G_t)$. Consequently, $s(t) = \cos(G_t, G_T)$ in Definition 1.3 is a quadratic sensitivity alignment measure distinct from $s_{\mathsf{NFA}}(t)$. All algebraic results developed below apply directly to $s_{\mathsf{NFA}}(t)$ if and only if G_t is replaced throughout by $M_t = G_t^{1/2}$ prior to window averaging and trace normalization $\tilde{M}_t := M_t/\mathsf{Tr}(M_t)$.

Definition 1.4 (Spectral Purity and Effective Rank)**.** Let $\{\lambda_j(\tilde{G}_t)\}_{j=1}^d$ denote the eigenvalues of $\tilde{G}_t$, which satisfy $\lambda_j(\tilde{G}_t) \geq 0$ and $\sum_{j=1}^d \lambda_j(\tilde{G}_t) = 1$. The **Spectral Purity** (inverse

participation ratio) $\mathcal{P}(t)$ and **Effective Rank** (participation ratio) $r_{\text{eff}}(t)$ of G_t are defined as:

$$\mathcal{P}(t) := \|\tilde{G}_t\|_F^2 = \mathsf{Tr}(\tilde{G}_t^2) = \frac{\|G_t\|_F^2}{\mathsf{Tr}(G_t)^2} = \sum_{j=1}^{d} \lambda_j(\tilde{G}_t)^2 \in \left[\frac{1}{d}, 1\right], \qquad r_{\text{eff}}(t) := \frac{1}{\mathcal{P}(t)} \in [1, d]. \quad (6)$$

Consistent with Definition 1.2, the baseline purity $\mathcal{P}_0$ and baseline effective rank d_{eff} are defined directly on the window-averaged reference operator $\tilde{G}_0$, and the asymptotic purity $\mathcal{P}_T$ and effective rank r are defined on $\tilde{G}_T$:

$$\mathcal{P}_0 := \|\tilde{G}_0\|_F^2 = \mathsf{Tr}(\tilde{G}_0^2) = \frac{1}{d_{\text{eff}}}, \qquad \mathcal{P}_T := \|\tilde{G}_T\|_F^2 = \mathsf{Tr}(\tilde{G}_T^2) = \frac{1}{r}, \quad (7)$$

where $r_{\text{eff},0} := d_{\text{eff}} = 1/\mathcal{P}_0$ is the participation ratio of $\tilde{G}_0$ and $r = 1/\mathcal{P}_T$ is the participation ratio of $\tilde{G}_T$.

2 Formal Modeling Hypotheses and Perturbation Structure

Intuition for the Reduced-Order Chord Model. In full generality, $\tilde{G}_t$ evolves in a $(d(d+1)/2 - 1)$-dimensional manifold. To infer alignment with the future state $\tilde{G}_T$ from scalar invariants at time t, the trajectory must be effectively low-dimensional after interpolation. We model the post-window evolution ($t \geq t_0$) as a two-state interpolation between the memorizing reference operator $\tilde{G}_0$ and the asymptotic generalizing operator $\tilde{G}_T$, and we explicitly parameterize the inner-product defect $\delta = \langle \tilde{G}_0, \tilde{G}_T - \tilde{G}_0 \rangle_F$ to track how non-isotropic structure at t_0 perturbs the geometry.

Assumption 2.1 (Two-State Spectraplex Chord Hypothesis)**.** For $t \geq t_0 = t_{\text{mem}} + w$, the trace-normalized AGOP $\tilde{G}_t$ evolves along the convex chord in $\Delta^{d\times d}$ connecting the reference operator $\tilde{G}_0$ and the asymptotic operator $\tilde{G}_T$:

$$\tilde{G}_t = (1 - \alpha_t)\tilde{G}_0 + \alpha_t \tilde{G}_T, \quad (8)$$

where $\alpha_t : [t_0, \infty) \to [\alpha_0, 1)$ is a continuous, monotonically increasing mixing function with initial anchor offset $\alpha_{t_0} = \alpha_0 \in [0, 1)$ (where $\alpha_0 = 0$ when G_τ is stationary over $[t_{\text{mem}}, t_0]$, i.e., $w \ll 1/\kappa$) and $\lim_{t\to\infty} \alpha_t = 1$ (so $\alpha_T \approx 1$ at any finite stopping horizon $T \gg t_0$).

Remark 2.1 (Scope and Theoretical Limitations of Assumption 2.1)*.* In unnormalized matrix space, Assumption 2.1 is equivalent to the conic containment $G_t \in \text{cone}\{G_0, G_T\}$, which states that G_t remains within the fixed two-dimensional subspace $\mathcal{V} := \text{span}\{G_0, G_T\} \subset \mathcal{S}^{d\times d}$ for $t \geq t_0$ and evolves purely by reweighting $\tilde{G}_0$ and $\tilde{G}_T$ without rotating intermediate eigenvectors outside $\mathcal{V}$. Even under an idealized linear weight decomposition $W_t = a_t W_0 + B_t$, a first-layer Gram matrix expands into three terms:

$$W_t^\top W_t = a_t^2 W_0^\top W_0 + a_t(W_0^\top B_t + B_t^\top W_0) + B_t^\top B_t,$$

and in multi-layer ReLU networks G_t couples all layer weights with input-dependent activation masks while the learned component B_t emerges gradually across the plateau. Moreover, by $t = t_0$ the network has already trained to interpolate the data, so $\tilde{G}_0$ is the post-interpolation operator rather than the raw random initialization kernel. Assumption 2.1 is therefore a secular two-state modelling hypothesis that neglects transient interference terms $W_0^\top B_t + B_t^\top W_0$ and intermediate subspace rotation; we provide a direct online test for it in Section 4. Finally, requiring α_t to be monotonically increasing restricts the model to single-stage rank compression and excludes non-monotonic late-collapse regimes.

Assumption 2.2 (Parameterized Travel-Direction Orthogonality Defect)**.** The Frobenius inner product between the reference operator $\tilde{G}_0$ and the displacement vector $\tilde{G}_T - \tilde{G}_0$ is parameterized by the orthogonality defect $\delta \in \mathbb{R}$:

$$\delta := \langle \tilde{G}_0, \tilde{G}_T - \tilde{G}_0 \rangle_F \iff \langle \tilde{G}_0, \tilde{G}_T \rangle_F = \mathcal{P}_0 + \delta. \tag{9}$$

When $\delta = 0$, the direction of travel $\tilde{G}_T - \tilde{G}_0$ is Frobenius-orthogonal to the starting operator $\tilde{G}_0$.

Proposition 2.1 (Structural Characterization of the Orthogonality Defect δ)**.** *Let $\tilde{G}_0, \tilde{G}_T \in \Delta^{d\times d}$ with $\mathcal{P}_0 = \mathsf{Tr}(\tilde{G}_0^2) = 1/d_{\mathit{eff}}$ and $\mathcal{P}_T = \mathsf{Tr}(\tilde{G}_T^2) = 1/r$.*

1. ***(Full Isotropic Baseline):** If $\tilde{G}_0 = \frac{1}{d} I_d$, then $d_{\mathit{eff}} = d$ and $\delta = 0$ identically.*
2. ***(Flat-Spectrum Subspace Projector):** If $\tilde{G}_0 = \frac{1}{k}\Pi_{\mathcal{U}}$, where $\Pi_{\mathcal{U}}$ is the orthogonal projector onto a subspace $\mathcal{U} \subseteq \mathbb{R}^d$ of dimension $k := \mathsf{dim}(\mathcal{U})$ containing the column space of $\tilde{G}_T$ ($\mathsf{col}(\tilde{G}_T) \subseteq \mathcal{U}$), then $d_{\mathit{eff}} = k$ and $\delta = 0$ identically.*
3. ***(Non-Flat Spectrum on $\mathcal{U}$ and Unaligned Eigenbasis Expectation):** Suppose $\mathsf{supp}(\tilde{G}_0) \subseteq \mathcal{U}$ and $\mathsf{col}(\tilde{G}_T) \subseteq \mathcal{U}$ with $k = \mathsf{dim}(\mathcal{U})$, where $\tilde{G}_0$ has a non-flat spectrum on $\mathcal{U}$ (so $d_{\mathit{eff}} < k$, i.e., $\mathcal{P}_0 > \frac{1}{k}$). Then for any deterministic realization:*

$$\left| \delta - \left(\frac{1}{k} - \mathcal{P}_0 \right) \right| \leq \sqrt{\mathcal{P}_0 - \frac{1}{k}} \sqrt{\mathcal{P}_T - \frac{1}{k}}. \tag{10}$$

Moreover, under the idealized modelling assumption that the eigenbasis U of $\tilde{G}_0 = U\Lambda_0 U^\top$ is distributed according to the Haar measure on $\mathcal{O}(\mathcal{U})$ independent of $\tilde{G}_T$ (an approximation, since in practice $\tilde{G}_T$ is learned starting from the state at t_0), $\mathbb{E}_U[\langle \tilde{G}_0, \tilde{G}_T \rangle_F] = \frac{1}{k} \neq \mathcal{P}_0$, yielding a strictly negative expected defect:

$$\mathbb{E}_U[\delta] = \frac{1}{k} - \mathcal{P}_0 = \frac{1}{\mathsf{dim}(\mathcal{U})} - \frac{1}{d_{\mathit{eff}}} < 0, \tag{11}$$

with equality $\mathbb{E}_U[\delta] = 0$ if and only if $\tilde{G}_0$ has a flat spectrum on $\mathcal{U}$ (reducing to Case 2).

Proof of Proposition 2.1. For Case 1, substituting $\tilde{G}_0 = \frac{1}{d} I_d$ gives:

$$\mathcal{P}_0 = \mathsf{Tr}\left(\left(\frac{1}{d} I_d \right)^2 \right) = \frac{1}{d^2} \mathsf{Tr}(I_d) = \frac{1}{d}.$$

Since $\mathsf{Tr}(\tilde{G}_T) = 1$, we have $\langle \tilde{G}_0, \tilde{G}_T \rangle_F = \frac{1}{d} \mathsf{Tr}(\tilde{G}_T) = \frac{1}{d} = \mathcal{P}_0$, which implies $\delta = \langle \tilde{G}_0, \tilde{G}_T \rangle_F - \mathcal{P}_0 = 0$.

For Case 2, since $\Pi_{\mathcal{U}}$ is an orthogonal projector onto a k-dimensional subspace, $\Pi_{\mathcal{U}}^2 = \Pi_{\mathcal{U}}$ and $\mathsf{Tr}(\Pi_{\mathcal{U}}) = k$. Thus:

$$\mathcal{P}_0 = \mathsf{Tr}\left(\left(\frac{1}{k} \Pi_{\mathcal{U}} \right)^2 \right) = \frac{1}{k^2} \mathsf{Tr}(\Pi_{\mathcal{U}}) = \frac{1}{k}.$$

Because $\mathsf{col}(\tilde{G}_T) \subseteq \mathcal{U}$, we have $\Pi_{\mathcal{U}} \tilde{G}_T = \tilde{G}_T$, and therefore $\langle \tilde{G}_0, \tilde{G}_T \rangle_F = \frac{1}{k} \mathsf{Tr}(\Pi_{\mathcal{U}} \tilde{G}_T) = \frac{1}{k} \mathsf{Tr}(\tilde{G}_T) = \frac{1}{k} = \mathcal{P}_0$, giving $\delta = 0$.

For Case 3, decompose $\tilde{G}_0$ and $\tilde{G}_T$ on $\mathcal{U}$ into the isotropic subspace baseline $\frac{1}{k}\Pi_{\mathcal{U}}$ and traceless deviators $\tilde{G}_0^\circ := \tilde{G}_0 - \frac{1}{k}\Pi_{\mathcal{U}}$ and $\tilde{G}_T^\circ := \tilde{G}_T - \frac{1}{k}\Pi_{\mathcal{U}}$. Because $\mathsf{Tr}(\tilde{G}_0^\circ) = 0$ and $\mathsf{Tr}(\tilde{G}_T^\circ) =$

0, the deviators are Frobenius-orthogonal to $\Pi_{\mathcal{U}}$, yielding $\|\tilde{G}_0^\circ\|_F^2 = \mathcal{P}_0 - \frac{1}{k}$ and $\|\tilde{G}_T^\circ\|_F^2 = \mathcal{P}_T - \frac{1}{k}$. Expanding the inner product gives:

$$\langle \tilde{G}_0, \tilde{G}_T \rangle_F = \frac{1}{k} + \langle \tilde{G}_0^\circ, \tilde{G}_T^\circ \rangle_F \implies \delta = \langle \tilde{G}_0, \tilde{G}_T \rangle_F - \mathcal{P}_0 = \left(\frac{1}{k} - \mathcal{P}_0 \right) + \langle \tilde{G}_0^\circ, \tilde{G}_T^\circ \rangle_F.$$

Applying the Cauchy-Schwarz inequality $|\langle \tilde{G}_0^\circ, \tilde{G}_T^\circ \rangle_F| \le \|\tilde{G}_0^\circ\|_F \|\tilde{G}_T^\circ\|_F$ establishes the deterministic bound. Under the Haar model $U \sim \text{Haar}(\mathcal{O}(\mathcal{U}))$ independent of $\tilde{G}_T$, rotational invariance on $\mathcal{U}$ yields $\mathbb{E}_U[U\Lambda_0 U^\top] = \frac{\text{Tr}(\Lambda_0)}{k}\Pi_{\mathcal{U}} = \frac{1}{k}\Pi_{\mathcal{U}}$, so $\mathbb{E}_U[\tilde{G}_0^\circ] = 0$ and $\mathbb{E}_U[\delta] = \frac{1}{k} - \mathcal{P}_0 < 0$ by the strict Cauchy-Schwarz inequality $\sum_{j=1}^k \lambda_j(\tilde{G}_0)^2 > \frac{1}{k}$ for any non-flat spectrum on $\mathcal{U}$. □

Assumption 2.3 (Strict Effective-Rank Compression)**.** Transitioning from the post-interpolation anchor ($t = t_0$) to asymptotic generalization ($t \to \infty$) strictly increases spectral purity (equivalently, strictly reduces effective rank), and the asymptotic operator is distinct from the reference operator:

$$\mathcal{P}_T > \mathcal{P}_0 \iff r < d_{\text{eff}}, \qquad \text{and} \qquad \Delta\mathcal{P}_\delta := \|\tilde{G}_T - \tilde{G}_0\|_F^2 = \mathcal{P}_T - \mathcal{P}_0 - 2\delta > 0. \tag{12}$$

3 Step-by-Step Derivation and Main Theorems

Intuition for Purity Evolution and the $\delta < 0$ Dead Zone. Expanding $\|\tilde{G}_t\|_F^2 = \|\tilde{G}_0 + \alpha_t(\tilde{G}_T - \tilde{G}_0)\|_F^2$ produces a quadratic function of α_t whose linear coefficient is $2\langle \tilde{G}_0, \tilde{G}_T - \tilde{G}_0 \rangle_F = 2\delta$. When $\delta = 0$, the linear term vanishes and purity grows purely quadratically as $\mathcal{P}_0 + \alpha_t^2(\mathcal{P}_T - \mathcal{P}_0)$, permitting immediate square-root inversion for α_t. However, when $\delta < 0$ (the realistic non-flat case identified in Proposition 2.1(3)), mixing $\tilde{G}_T$ into $\tilde{G}_0$ initially *flattens* the spectrum before concentrating it, causing $\mathcal{P}(t)$ to dip below $\mathcal{P}_0$ on an initial interval $\alpha_t \in (0, -2\delta/\Delta\mathcal{P}_\delta)$. Over this "dead zone," purity $\mathcal{P}(t)$ is below $\mathcal{P}_0$, so unique inversion of α_t from $\mathcal{P}(t)$ requires $\mathcal{P}(t) \ge \mathcal{P}_0$.

Lemma 3.1 (Exact Purity Evolution, Unique Root Condition, and the $\delta < 0$ Dead Zone)**.** *Under Assumptions 2.1–2.3, the spectral purity $\mathcal{P}(t)$ at any $t \ge t_0$ satisfies the exact quadratic law:*

$$\mathcal{P}(t) = \mathcal{P}_0 + 2\alpha_t\delta + \alpha_t^2\Delta\mathcal{P}_\delta = \mathcal{P}_0 + 2\alpha_t\delta + \alpha_t^2(\mathcal{P}_T - \mathcal{P}_0 - 2\delta). \tag{13}$$

Consequently:

1. ***(Unique Non-Negative Root for $\mathcal{P}(t) \ge \mathcal{P}_0$):*** *Whenever $\mathcal{P}(t) > \mathcal{P}_0$ (or $\mathcal{P}(t) \ge \mathcal{P}_0$ with $\delta \ge 0$), Equation (13) admits a unique non-negative root $\alpha_t \in [0, 1)$, given by:*

$$\alpha_t = \frac{-\delta + \sqrt{\delta^2 + (\mathcal{P}(t) - \mathcal{P}_0)\Delta\mathcal{P}_\delta}}{\Delta\mathcal{P}_\delta}. \tag{14}$$

   In the orthogonal-travel regime ($\delta = 0$), $\mathcal{P}(t) \ge \mathcal{P}_0$ holds for all $t \ge t_0$, and Equations (13)–(14) reduce to:

$$\mathcal{P}(t) = \mathcal{P}_0 + \alpha_t^2(\mathcal{P}_T - \mathcal{P}_0), \qquad \alpha_t = \sqrt{\frac{\mathcal{P}(t) - \mathcal{P}_0}{\mathcal{P}_T - \mathcal{P}_0}}. \tag{15}$$

2. ***(The $\delta < 0$ Purity Dead Zone):*** *If $\delta < 0$, then $\left.\frac{d\mathcal{P}}{d\alpha_t}\right|_{\alpha_t=0} = 2\delta < 0$. As α_t increases from 0, $\mathcal{P}(t)$ initially drops below $\mathcal{P}_0$, attaining a global minimum $\mathcal{P}_{\min} = \mathcal{P}_0 - \frac{\delta^2}{\Delta\mathcal{P}_\delta}$ at $\alpha_t = \frac{-\delta}{\Delta\mathcal{P}_\delta}$, and strictly recovers to $\mathcal{P}(t) > \mathcal{P}_0$ only after α_t crosses the dead-zone threshold:*

$$\alpha_t > \alpha_{\textit{dead}} := \frac{-2\delta}{\Delta\mathcal{P}_\delta} \in (0, 1). \tag{16}$$

For $\alpha_t \in (0, \alpha_{dead})$, $\mathcal{P}(t) < \mathcal{P}_0$: both roots of $\Delta\mathcal{P}_\delta \alpha^2 + 2\delta\alpha - (\mathcal{P}(t) - \mathcal{P}_0) = 0$ are positive, so α_t cannot be uniquely recovered from $\mathcal{P}(t)$ alone without tracking passage past $\mathcal{P}_{\text{min}}$, and any estimator requiring $\mathcal{P}(t) \geq \mathcal{P}_0$ is undefined on this interval.

Proof of Lemma 3.1. Rewrite Assumption 2.1 in displacement form as $\tilde{G}_t = \tilde{G}_0 + \alpha_t(\tilde{G}_T - \tilde{G}_0)$. Expanding $\mathcal{P}(t) = \|\tilde{G}_t\|_F^2$ by bilinearity of the Frobenius inner product gives:

$$\begin{aligned}\mathcal{P}(t) &= \left\langle \tilde{G}_0 + \alpha_t(\tilde{G}_T - \tilde{G}_0),\, \tilde{G}_0 + \alpha_t(\tilde{G}_T - \tilde{G}_0) \right\rangle_F \\ &= \|\tilde{G}_0\|_F^2 + 2\alpha_t \langle \tilde{G}_0, \tilde{G}_T - \tilde{G}_0 \rangle_F + \alpha_t^2 \|\tilde{G}_T - \tilde{G}_0\|_F^2.\end{aligned}$$

By Definition 1.4 and Assumption 2.2, $\|\tilde{G}_0\|_F^2 = \mathcal{P}_0$ and $\langle \tilde{G}_0, \tilde{G}_T - \tilde{G}_0 \rangle_F = \delta$. Expanding the squared displacement norm yields:

$$\|\tilde{G}_T - \tilde{G}_0\|_F^2 = \|\tilde{G}_T\|_F^2 - 2\langle \tilde{G}_0, \tilde{G}_T \rangle_F + \|\tilde{G}_0\|_F^2 = \mathcal{P}_T - 2(\mathcal{P}_0 + \delta) + \mathcal{P}_0 = \mathcal{P}_T - \mathcal{P}_0 - 2\delta = \Delta\mathcal{P}_\delta.$$

This yields the quadratic equation $\Delta\mathcal{P}_\delta \alpha^2 + 2\delta\alpha - (\mathcal{P}(t) - \mathcal{P}_0) = 0$ with roots $\alpha_\pm = \frac{-\delta \pm \sqrt{\delta^2 + (\mathcal{P}(t) - \mathcal{P}_0)\Delta\mathcal{P}_\delta}}{\Delta\mathcal{P}_\delta}$. When $\mathcal{P}(t) > \mathcal{P}_0$, the product of the roots is $\alpha_+\alpha_- = -\frac{\mathcal{P}(t) - \mathcal{P}_0}{\Delta\mathcal{P}_\delta} < 0$ (since $\Delta\mathcal{P}_\delta > 0$ by Assumption 2.3), so $\alpha_- < 0 < \alpha_+$ and $\alpha_+ = \alpha_t$ is the unique positive root. When $\delta < 0$, factoring $\mathcal{P}(t) - \mathcal{P}_0 = \alpha_t(2\delta + \alpha_t \Delta\mathcal{P}_\delta)$ shows that $\mathcal{P}(t) < \mathcal{P}_0$ precisely for $\alpha_t \in (0, -2\delta/\Delta\mathcal{P}_\delta)$ (where $-2\delta/\Delta\mathcal{P}_\delta < 1$ because $\Delta\mathcal{P}_\delta + 2\delta = \mathcal{P}_T - \mathcal{P}_0 > 0$). On this interval, $\alpha_+\alpha_- > 0$ and $\alpha_+ + \alpha_- = -2\delta/\Delta\mathcal{P}_\delta > 0$, so both roots $\alpha_\pm$ are positive. □

Lemma 3.2 (Cross-Inner Products with Reference and Asymptotic Operators)**.** *Under Assumptions 2.1–2.3, for all $t \geq t_0$:*

1. *The inner product with the reference operator $\tilde{G}_0$ is always non-negative (as the Frobenius inner product of two PSD matrices) and satisfies:*

$$\langle \tilde{G}_t, \tilde{G}_0 \rangle_F = \mathcal{P}_0 + \alpha_t \delta \geq 0 \implies \cos(G_t, G_0) = \frac{\mathcal{P}_0 + \alpha_t \delta}{\sqrt{\mathcal{P}(t)\mathcal{P}_0}}. \tag{17}$$

   When $\delta = 0$, $\langle \tilde{G}_t, \tilde{G}_0 \rangle_F = \mathcal{P}_0$ is invariant in time and $\cos(G_t, G_0) = \sqrt{\frac{\mathcal{P}_0}{\mathcal{P}(t)}}$.

2. *The inner product with the asymptotic operator $\tilde{G}_T$ satisfies:*

$$\langle \tilde{G}_t, \tilde{G}_T \rangle_F = \mathcal{P}_0 + \delta + \alpha_t(\mathcal{P}_T - \mathcal{P}_0 - \delta). \tag{18}$$

   When $\delta = 0$, substituting α_t from Lemma 3.1 yields:

$$\langle \tilde{G}_t, \tilde{G}_T \rangle_F = \mathcal{P}_0 + \sqrt{(\mathcal{P}(t) - \mathcal{P}_0)(\mathcal{P}_T - \mathcal{P}_0)}. \tag{19}$$

Proof of Lemma 3.2. For Part 1, since $\tilde{G}_t, \tilde{G}_0 \in \mathcal{S}_+^{d\times d}$, $\langle \tilde{G}_t, \tilde{G}_0 \rangle_F = \text{Tr}(\tilde{G}_0^{1/2} \tilde{G}_t \tilde{G}_0^{1/2}) \geq 0$. Taking the inner product of $\tilde{G}_t = \tilde{G}_0 + \alpha_t(\tilde{G}_T - \tilde{G}_0)$ with $\tilde{G}_0$ gives $\langle \tilde{G}_t, \tilde{G}_0 \rangle_F = \mathcal{P}_0 + \alpha_t\delta$, and dividing by $\|\tilde{G}_t\|_F \|\tilde{G}_0\|_F = \sqrt{\mathcal{P}(t)\mathcal{P}_0}$ gives $\cos(G_t, G_0)$.

For Part 2, taking the inner product of $\tilde{G}_t = (1 - \alpha_t)\tilde{G}_0 + \alpha_t \tilde{G}_T$ with $\tilde{G}_T$ and applying Assumption 2.2 gives:

$$\langle \tilde{G}_t, \tilde{G}_T \rangle_F = (1-\alpha_t)\langle \tilde{G}_0, \tilde{G}_T \rangle_F + \alpha_t \|\tilde{G}_T\|_F^2 = (1-\alpha_t)(\mathcal{P}_0 + \delta) + \alpha_t \mathcal{P}_T = \mathcal{P}_0 + \delta + \alpha_t(\mathcal{P}_T - \mathcal{P}_0 - \delta).$$

Setting $\delta = 0$ and substituting $\alpha_t = \sqrt{\frac{\mathcal{P}(t) - \mathcal{P}_0}{\mathcal{P}_T - \mathcal{P}_0}}$ from Equation (15) gives $\langle \tilde{G}_t, \tilde{G}_T \rangle_F = \mathcal{P}_0 + \sqrt{(\mathcal{P}(t) - \mathcal{P}_0)(\mathcal{P}_T - \mathcal{P}_0)}$. □

Definition 3.1 (Online Spiked-AGOP Purity Estimator and Projection Estimator)**.** For any $t \geq t_0$, we define two causal observables computable online at time t:

1. The **Online Spiked-AGOP Purity Estimator** $\widehat{s}_{\text{spiked}}(t)$, defined whenever $\mathcal{P}(t) \geq \mathcal{P}_0$ (i.e., outside the $\delta < 0$ dead zone $\alpha_t \in (0, \alpha_{\text{dead}})$ of Lemma 3.1(2)):

$$\widehat{s}_{\text{spiked}}(t) := \sqrt{1 - \frac{\mathcal{P}_0}{\mathcal{P}(t)}} = \sqrt{1 - \frac{\|\tilde{G}_0\|_F^2 \, \text{Tr}(G_t)^2}{\|G_t\|_F^2}}. \tag{20}$$

2. The **Online Baseline-Projection Estimator** $\widehat{s}_{\text{proj}}(t)$, which is well-defined for all $t \geq t_0$ (including inside the $\delta < 0$ dead zone) and measures the sine of the Frobenius angle between G_t and the reference anchor G_0:

$$\widehat{s}_{\text{proj}}(t) := \sqrt{1 - \cos^2(G_t, G_0)}. \tag{21}$$

Proposition 3.3 (Identity as Fractional Drop in Effective Rank)**.** *By Definition 1.4 ($r_{\text{eff}}(t) = 1/\mathcal{P}(t)$ and $r_{\text{eff},0} = d_{\text{eff}} = 1/\mathcal{P}_0$), whenever $\mathcal{P}(t) \geq \mathcal{P}_0$, the squared online estimator $\widehat{s}_{\text{spiked}}(t)^2$ is identically the fractional reduction in effective rank relative to the reference operator $\tilde{G}_0$:*

$$\widehat{s}_{\text{spiked}}(t)^2 = 1 - \frac{\mathcal{P}_0}{\mathcal{P}(t)} = 1 - \frac{r_{\text{eff}}(t)}{r_{\text{eff},0}}. \tag{22}$$

Thus, $\widehat{s}_{\text{spiked}}(t)$ is an effective-rank drop statistic that functions as a formal alignment proxy under Assumptions 2.1–2.3.

Theorem 3.4 (Exact Law Relating Online Estimators and Offline Alignment)**.** *Because $\tilde{G}_0, \tilde{G}_T \in \mathcal{S}_+^{d\times d}$ are PSD and linearly independent, their initial alignment satisfies $s_{\text{init}} := \cos(G_0, G_T) = \frac{\mathcal{P}_0 + \delta}{\sqrt{\mathcal{P}_0 \mathcal{P}_T}} \in [0, 1)$, and its $\delta = 0$ value is $s_0 := \sqrt{\frac{\mathcal{P}_0}{\mathcal{P}_T}} = \sqrt{\frac{r}{d_{\text{eff}}}} \in (0, 1)$.*

1. ***(Exactness for $\widehat{s}_{\text{proj}}(t)$ under Assumption 2.1 alone):*** *For any orthogonality defect δ, under Assumption 2.1 alone, the offline alignment $s(t)$ satisfies the exact planar angle-addition law for all $t \geq t_0$:*

$$s(t) = \widehat{s}_{\text{proj}}(t)\sqrt{1 - s_{\text{init}}^2} + s_{\text{init}}\sqrt{1 - \widehat{s}_{\text{proj}}(t)^2}. \tag{23}$$

2. ***(Exactness for $\widehat{s}_{\text{spiked}}(t)$ under $\delta = 0$):*** *Under Assumptions 2.1–2.3 with $\delta = 0$, $\widehat{s}_{\text{proj}}(t) = \widehat{s}_{\text{spiked}}(t)$ and $s_{\text{init}} = s_0$, so that for all $t \geq t_0$:*

$$s(t) = \widehat{s}_{\text{spiked}}(t)\sqrt{1 - s_0^2} + s_0\sqrt{1 - \widehat{s}_{\text{spiked}}(t)^2} = \sin\big(\psi(t) + \theta_0\big), \tag{24}$$

   where $\psi(t) := \arcsin\big(\widehat{s}_{\text{spiked}}(t)\big) \in [0, \frac{\pi}{2} - \theta_0)$ and $\theta_0 := \arcsin(s_0) \in (0, \frac{\pi}{2})$.

3. ***(Perturbation Bound for Uncalibrated $\widehat{s}_{\text{spiked}}(t)$ when $\delta \neq 0$):*** *Let $s^{\text{law}}_{\text{spiked}}(t) := \sin\big(\psi(t) + \theta_0\big)$ denote the uncalibrated $\delta = 0$ reconstruction formula of Equation (24), and let $\theta_{\text{init}} := \arcsin(s_{\text{init}}) \in [0, \frac{\pi}{2})$ and $\psi_{\text{proj}}(t) := \arcsin\big(\widehat{s}_{\text{proj}}(t)\big)$. For $\delta \geq 0$ (where $\mathcal{P}(t) \geq \mathcal{P}_0$ for all $t \geq t_0$) and $\theta_{\max} := \max(\theta_0, \theta_{\text{init}}) < \frac{\pi}{2}$:*

$$\sup_{t\geq t_0} \big|s(t) - s^{\text{law}}_{\text{spiked}}(t)\big| \leq |\theta_{\text{init}} - \theta_0| + \sup_{t\geq t_0} |\psi_{\text{proj}}(t) - \psi(t)| \leq \frac{|\delta|}{\cos\theta_{\max}\sqrt{\mathcal{P}_0\mathcal{P}_T}} + \frac{2|\delta|}{\sqrt{\mathcal{P}_0 \Delta\mathcal{P}_\delta}} + \mathcal{O}(\delta^2). \tag{25}$$

   In particular, whenever $\Delta\mathcal{P}_\delta \geq \mathcal{P}_0$, the second term is bounded by $\mathcal{O}(|\delta|/\mathcal{P}_0)$. For the negative-defect regime $\delta < 0$ (Proposition 2.1(3)), $s^{\text{law}}_{\text{spiked}}(t)$ is undefined inside the dead zone $\alpha_t \in (0, \alpha_{\text{dead}})$ and no bound is claimed for $\delta < 0$; this regime is handled directly by the exact $\delta \neq 0$ self-calibrated reconstruction $\widehat{s}_{\text{cal},\delta}(t)$ in Theorem 5.2(2).

Proof of Theorem 3.4. For Part 1, under Assumption 2.1, $\tilde{G}_t = (1-\alpha_t)\tilde{G}_0 + \alpha_t \tilde{G}_T$ with $\alpha_t \in [0,1)$ lies in the two-dimensional Euclidean subspace $\mathcal{V} = \mathrm{span}\{\tilde{G}_0, \tilde{G}_T\}$ within the positive planar sector bounded by $\tilde{G}_0$ and $\tilde{G}_T$. Let $\Phi_{0,T} := \arccos(\cos(G_0, G_T)) = \frac{\pi}{2} - \arcsin(s_{\text{init}})$ be the Frobenius angle between G_0 and G_T, and let $\psi_{\text{proj}}(t) := \arccos(\cos(G_t, G_0)) = \arcsin(\widehat{s}_{\text{proj}}(t)) \in [0, \Phi_{0,T})$ be the Frobenius angle between G_0 and G_t. Because G_t lies in the positive planar sector between G_0 and G_T, the angle between G_t and G_T is $\Phi_{0,T} - \psi_{\text{proj}}(t) = \frac{\pi}{2} - (\psi_{\text{proj}}(t) + \arcsin(s_{\text{init}}))$. Taking the cosine yields:

$$s(t) = \cos\Big(\frac{\pi}{2} - \big(\psi_{\text{proj}}(t) + \arcsin(s_{\text{init}})\big)\Big) = \sin\big(\psi_{\text{proj}}(t) + \arcsin(s_{\text{init}})\big).$$

Expanding via the trigonometric identity $\sin(A+B) = \sin A \cos B + \cos A \sin B$ proves Equation (23).

For Part 2, when $\delta = 0$, dividing $\langle \tilde{G}_t, \tilde{G}_T \rangle_F = \mathcal{P}_0 + \sqrt{(\mathcal{P}(t) - \mathcal{P}_0)(\mathcal{P}_T - \mathcal{P}_0)}$ from Lemma 3.2(2) by $\|\tilde{G}_t\|_F \|\tilde{G}_T\|_F = \sqrt{\mathcal{P}(t)\mathcal{P}_T}$ gives:

$$s(t) = \sqrt{\frac{\mathcal{P}_0}{\mathcal{P}(t)}}\sqrt{\frac{\mathcal{P}_0}{\mathcal{P}_T}} + \sqrt{1 - \frac{\mathcal{P}_0}{\mathcal{P}(t)}}\sqrt{1 - \frac{\mathcal{P}_0}{\mathcal{P}_T}} = s_0\sqrt{1 - \widehat{s}_{\text{spiked}}(t)^2} + \widehat{s}_{\text{spiked}}(t)\sqrt{1 - s_0^2} = \sin\big(\psi(t) + \theta_0\big).$$

For Part 3, since $x \mapsto \sin(x)$ is 1-Lipschitz, $|s(t) - s^{\text{law}}_{\text{spiked}}(t)| = |\sin(\psi_{\text{proj}}(t) + \theta_{\text{init}}) - \sin(\psi(t) + \theta_0)| \le |\theta_{\text{init}} - \theta_0| + |\psi_{\text{proj}}(t) - \psi(t)|$. For the first term, applying the Mean Value Theorem to arcsin on $[s_0, s_{\text{init}}]$ yields:

$$|\theta_{\text{init}} - \theta_0| = |\arcsin(s_{\text{init}}) - \arcsin(s_0)| \le \frac{|s_{\text{init}} - s_0|}{\sqrt{1 - \sin^2\theta_{\max}}} = \frac{|\delta|}{\cos\theta_{\max}\sqrt{\mathcal{P}_0\mathcal{P}_T}}.$$

For the second term, notice that arccos has unbounded derivative near 1 (at $t \approx t_0$ where $\psi \approx 0$), so we instead pass through arctan, which is globally 1-Lipschitz on $[0,\infty)$ ($\frac{d}{dx}\arctan(x) = \frac{1}{1+x^2} \le 1$). From Lemma 3.1 and Lemma 3.2(1), we have $\cos\psi(t) = \sqrt{\mathcal{P}_0/\mathcal{P}(t)}$ and $\cos\psi_{\text{proj}}(t) = (\mathcal{P}_0 + \alpha_t\delta)/\sqrt{\mathcal{P}_0\mathcal{P}(t)}$. Using $\mathcal{P}_0\mathcal{P}(t) - (\mathcal{P}_0 + \alpha_t\delta)^2 = \alpha_t^2(\mathcal{P}_0\Delta\mathcal{P}_\delta - \delta^2)$, the tangents of $\psi(t)$ and $\psi_{\text{proj}}(t)$ are:

$$\tan\psi(t) = \sqrt{\frac{2\alpha_t\delta + \alpha_t^2\Delta\mathcal{P}_\delta}{\mathcal{P}_0}} = \sqrt{x^2 + 2\eta x}, \qquad \tan\psi_{\text{proj}}(t) = \frac{\alpha_t\sqrt{\mathcal{P}_0\Delta\mathcal{P}_\delta - \delta^2}}{\mathcal{P}_0 + \alpha_t\delta} = \frac{x\sqrt{1-\eta^2}}{1+\eta x},$$

where $x := \alpha_t\sqrt{\Delta\mathcal{P}_\delta/\mathcal{P}_0} \ge 0$ and $\eta := \delta/\sqrt{\mathcal{P}_0\Delta\mathcal{P}_\delta} \ge 0$. Using the exact tangent-difference formula $\tan(\psi(t) - \psi_{\text{proj}}(t)) = \frac{\tan\psi(t) - \tan\psi_{\text{proj}}(t)}{1 + \tan\psi(t)\tan\psi_{\text{proj}}(t)}$ gives:

$$\big|\psi(t) - \psi_{\text{proj}}(t)\big| \le \big|\tan\big(\psi(t) - \psi_{\text{proj}}(t)\big)\big| \le \big(\sqrt{x^2+2\eta x} - x\big) + \frac{x - \frac{x\sqrt{1-\eta^2}}{1+\eta x}}{1 + \sqrt{x^2+2\eta x}\,\frac{x\sqrt{1-\eta^2}}{1+\eta x}}.$$

Rationalizing the first term gives $\sqrt{x^2+2\eta x} - x = \frac{2\eta x}{\sqrt{x^2+2\eta x}+x} = \frac{2\eta}{\sqrt{1+2\eta/x}+1} \le \eta$ for all $x \ge 0$, and the second term is bounded by $\frac{\eta x^2}{1+x^2} + \mathcal{O}(\eta^2) \le \eta + \mathcal{O}(\eta^2)$ uniformly over $x \ge 0$. Substituting $\eta = \frac{|\delta|}{\sqrt{\mathcal{P}_0\Delta\mathcal{P}_\delta}}$ completes the proof. □

Theorem 3.5 (Exact Subspace Interpretation: Baseline-Orthogonalized Terminal Direction)**.** *Let $\hat{G}_0 := \frac{\tilde{G}_0}{\|\tilde{G}_0\|_F}$ and $\hat{G}_T := \frac{\tilde{G}_T}{\|\tilde{G}_T\|_F}$ denote the unit Frobenius normalizations of $\tilde{G}_0$ and $\tilde{G}_T$. Define the* ***baseline-orthogonalized terminal direction*** *$\hat{G}_T^\perp \in \mathcal{S}^{d\times d}$ via Gram-Schmidt orthogonalization of $\hat{G}_T$ against $\hat{G}_0$:*

$$\hat{G}_T^\perp := \frac{\hat{G}_T - \langle \hat{G}_T, \hat{G}_0\rangle_F \hat{G}_0}{\|\hat{G}_T - \langle \hat{G}_T, \hat{G}_0\rangle_F \hat{G}_0\|_F}. \tag{26}$$

Then:

1. *Under Assumption 2.1 alone (for any δ), $\widehat{s}_{proj}(t) = \cos(G_t, \hat{G}_T^{\perp})$ for all $t \geq t_0$.*

2. *Under Assumptions 2.1–2.3 with $\delta = 0$, $\widehat{s}_{spiked}(t) = \cos(G_t, \hat{G}_T^{\perp})$ identically for all $t \geq t_0$ without requiring $r \ll d_{eff}$.*

Proof of Theorem 3.5. By Gram-Schmidt construction, $\{\hat{G}_0, \hat{G}_T^{\perp}\}$ forms an orthonormal basis for $\mathcal{V} = \text{span}\{\tilde{G}_0, \tilde{G}_T\}$ with $\langle \hat{G}_T, \hat{G}_T^{\perp} \rangle_F = \sqrt{1 - \langle \hat{G}_T, \hat{G}_0 \rangle_F^2} > 0$. By Assumption 2.1, $\hat{G}_t := \tilde{G}_t / \|\tilde{G}_t\|_F \in \mathcal{V}$ has non-negative projection onto $\hat{G}_T^{\perp}$ because:

$$\langle \tilde{G}_t, \hat{G}_T^{\perp} \rangle_F = (1 - \alpha_t)\langle \tilde{G}_0, \hat{G}_T^{\perp} \rangle_F + \alpha_t \langle \tilde{G}_T, \hat{G}_T^{\perp} \rangle_F = \alpha_t \|\tilde{G}_T\|_F \langle \hat{G}_T, \hat{G}_T^{\perp} \rangle_F \geq 0.$$

Applying Parseval's identity on $\mathcal{V}$ gives:

$$1 = \|\hat{G}_t\|_F^2 = \langle \hat{G}_t, \hat{G}_0 \rangle_F^2 + \langle \hat{G}_t, \hat{G}_T^{\perp} \rangle_F^2 = \cos^2(G_t, G_0) + \cos^2(G_t, \hat{G}_T^{\perp}).$$

Solving for $\cos(G_t, \hat{G}_T^{\perp}) \geq 0$ yields $\cos(G_t, \hat{G}_T^{\perp}) = \sqrt{1 - \cos^2(G_t, G_0)} = \widehat{s}_{\text{proj}}(t)$, proving Part 1. When $\delta = 0$, Lemma 3.2(1) gives $\cos^2(G_t, G_0) = \frac{\mathcal{P}_0}{\mathcal{P}(t)}$, so $\widehat{s}_{\text{proj}}(t) = \sqrt{1 - \frac{\mathcal{P}_0}{\mathcal{P}(t)}} = \widehat{s}_{\text{spiked}}(t)$, proving Part 2. □

Remark 3.1 (Indefiniteness of $\hat{G}_T^{\perp}$). Unlike $\tilde{G}_0$ and $\tilde{G}_T$, the matrix $\hat{G}_T^{\perp}$ is generally **not** positive semi-definite ($\hat{G}_T^{\perp} \notin \mathcal{S}_+^{d \times d}$). In the isotropic baseline case $\tilde{G}_0 = \frac{1}{d} I_d$ ($\delta = 0$), the unnormalized difference $\tilde{G}_T - \frac{1}{d} I_d$ is traceless and takes strictly negative eigenvalues $-\frac{1}{d}$ on $\ker(\tilde{G}_T)$; consequently, the unit-normalized matrix $\hat{G}_T^{\perp} = (\tilde{G}_T - \frac{1}{d} I_d)/\sqrt{\mathcal{P}_T - 1/d}$ takes strictly negative eigenvalues $-\frac{1}{d\sqrt{\mathcal{P}_T - 1/d}} < 0$ on $\ker(\tilde{G}_T)$. Thus, $\hat{G}_T^{\perp}$ represents a tangent direction on the spectraplex rather than a PSD covariance operator.

Corollary 3.6 (Order Isomorphism, Non-Asymptotic Error Bound, and Raw Thresholding Limitation)**.** *Under Assumptions 2.1–2.3 with $\delta = 0$:*

1. ***(Order Isomorphism):*** *$s(t) = \sin(\arcsin(\widehat{s}_{spiked}(t)) + \theta_0)$ is a strictly increasing bijection of $\widehat{s}_{spiked}(t)$ mapping $[\widehat{s}_{spiked}(t_0), \sqrt{1 - r/d_{eff}})$ onto $[s(t_0), 1)$ (which equals $[0, \sqrt{1 - r/d_{eff}}) \to [\sqrt{r/d_{eff}}, 1)$ when $\alpha_0 = 0$).*

2. ***(Tight Uniform Discrepancy):*** *The supremum discrepancy between $s(t)$ and $\widehat{s}_{spiked}(t)$ over $\alpha_t \in [0, 1)$ is attained at $\alpha_t = 0$ (i.e., at $t = t_0$ when $\alpha_0 = 0$, and is strictly smaller over $t \geq t_0$ when $\alpha_0 > 0$) and satisfies:*

$$\sup_{t \geq t_0} \left| s(t) - \widehat{s}_{spiked}(t) \right| \leq \sqrt{\frac{r}{d_{eff}}}, \tag{27}$$

   with equality when $\alpha_0 = 0$.

3. ***(Unreachability of Fixed Near-Unity Thresholds):*** *Because $\lim_{t \to \infty} \widehat{s}_{spiked}(t) = \sqrt{1 - \frac{r}{d_{eff}}} < 1$, any fixed stopping rule of the form $\widehat{s}_{spiked}(t) \geq 1 - \epsilon$ is strictly unreachable whenever the unknown asymptotic effective rank satisfies $r \geq d_{eff}(2\epsilon - \epsilon^2)$.*

Proof of Corollary 3.6. Parts 1 and 3 follow immediately from Theorem 3.4(2) because $\psi(t) \in [\psi(t_0), \frac{\pi}{2} - \theta_0) \subseteq [0, \frac{\pi}{2} - \theta_0)$ and $\lim_{t \to \infty} \widehat{s}_{\text{spiked}}(t) = \cos\theta_0 = \sqrt{1 - r/d_{\text{eff}}}$. For Part 2, applying the trigonometric identity $\sin(\psi + \theta_0) - \sin\psi = 2\sin(\frac{\theta_0}{2})\cos(\psi + \frac{\theta_0}{2})$ over $\psi \in [0, \frac{\pi}{2} - \theta_0)$ shows that $s(t) - \widehat{s}_{\text{spiked}}(t)$ is positive and strictly decreasing in ψ, attaining its maximum at $\psi = 0$ ($\alpha_t = 0$, which corresponds to $t = t_0$ when $\alpha_0 = 0$), where $2\sin(\frac{\theta_0}{2})\cos(\frac{\theta_0}{2}) = \sin\theta_0 = s_0 = \sqrt{r/d_{\text{eff}}}$. □

4 Online Diagnostics for Chord Collinearity and Orthogonality Drift

Intuition for Separating the Two Model Assumptions. Comparing $\widehat{s}_{\text{proj}}(t)^2$ and $\widehat{s}_{\text{spiked}}(t)^2$ tests only a single scalar projection along $\tilde{G}_0$: any off-chord residual R_t that happens to be orthogonal to $\tilde{G}_0$ ($\langle R_t, \tilde{G}_0\rangle_F = 0$) leaves that scalar comparison unchanged. To test Assumption 2.1 and Assumption 2.2 independently online, we observe two geometric facts:

1. Under Assumption 2.1 alone, $\tilde{G}_t - \tilde{G}_0 = \alpha_t(\tilde{G}_T - \tilde{G}_0)$ lies along a fixed ray from $\tilde{G}_0$ for all $t > t_0$. Therefore, consecutive displacement vectors $\tilde{G}_t - \tilde{G}_0$ and $\tilde{G}_{t-\Delta} - \tilde{G}_0$ must be collinear (cosine similarity 1), regardless of δ.

2. Under Assumptions 2.1 and 2.2, the projection of the displacement $\tilde{G}_t - \tilde{G}_0$ onto $\tilde{G}_0$ is the linear observable $v(t) := \langle \tilde{G}_t, \tilde{G}_0\rangle_F - \mathcal{P}_0 = \alpha_t\delta$, which directly tracks the sign and magnitude of the orthogonality defect δ.

Theorem 4.1 (Online Chord Collinearity Diagnostic and Linear Orthogonality-Drift Observable)**.** *For any arbitrary trace-normalized trajectory $\tilde{G}_t \in \Delta^{d\times d}$ ($t \geq t_0$), define the latent analytical projection coefficient α_t and out-of-chord residual R_t via orthogonal projection of the displacement $\tilde{G}_t - \tilde{G}_0$ onto the chord direction $\tilde{G}_T - \tilde{G}_0$:*

$$\alpha_t := \frac{\langle \tilde{G}_t - \tilde{G}_0,\, \tilde{G}_T - \tilde{G}_0\rangle_F}{\|\tilde{G}_T - \tilde{G}_0\|_F^2}, \qquad R_t := (\tilde{G}_t - \tilde{G}_0) - \alpha_t(\tilde{G}_T - \tilde{G}_0), \tag{28}$$

so that $\tilde{G}_t = (1-\alpha_t)\tilde{G}_0 + \alpha_t\tilde{G}_T + R_t$ where $R_t \in \mathcal{S}^{d\times d}$ is uniquely defined and satisfies $\langle R_t, \tilde{G}_T - \tilde{G}_0\rangle_F = 0$ and $\mathrm{Tr}(R_t) = 0$. (Note that α_t and R_t depend on $\tilde{G}_T$ and serve strictly as latent variables for theoretical analysis, whereas $D_{\text{chord}}(t)$, $v(t)$, and $\Delta_{\text{model}}(t)$ below are causal observables computable directly online at time t.) Let $\delta = \langle \tilde{G}_0, \tilde{G}_T - \tilde{G}_0\rangle_F$.

1. ***(Direct Test of Assumption 2.1 via Displacement Collinearity):** For any lag $\Delta > 0$ and $t > t_0 + \Delta$ with $\tilde{G}_t \neq \tilde{G}_0$ and $\tilde{G}_{t-\Delta} \neq \tilde{G}_0$, define the **Online Chord Collinearity Defect**:*

$$D_{\text{chord}}(t) := 1 - \cos\left(\tilde{G}_t - \tilde{G}_0,\, \tilde{G}_{t-\Delta} - \tilde{G}_0\right) \in [0, 2]. \tag{29}$$

Under Assumption 2.1 ($R_t \equiv 0$, $\alpha_t > 0$), $D_{\text{chord}}(t) = 0$ identically for all $t > t_0 + \Delta$ and any $\delta \in \mathbb{R}$. Conversely, $D_{\text{chord}}(t) > 0$ if and only if the displacement $\tilde{G}_t - \tilde{G}_0$ rotates out of the ray spanned by $\tilde{G}_{t-\Delta} - \tilde{G}_0$. (To prevent mini-batch noise from dominating the cosine when $\tilde{G}_t - \tilde{G}_0$ is small near t_0, in practice $D_{\text{chord}}(t)$ is evaluated on window-averaged operators $\bar{G}_t - \tilde{G}_0$ and $\bar{G}_{t-\Delta} - \tilde{G}_0$ with lag $\Delta \gtrsim 1/\widehat{\kappa}$ after a minimum burn-in $\widehat{\alpha}_t \geq \alpha_{\min} > 0$.)

2. ***(Linear Orthogonality-Drift Observable $v(t)$ and Relation to $\Delta_{\text{model}}(t)$):** Define the **Linear Baseline-Drift Observable** $v(t)$ and the **Squared Estimator Gap** $\Delta_{\text{model}}(t)$:*

$$v(t) := \langle \tilde{G}_t, \tilde{G}_0\rangle_F - \mathcal{P}_0, \qquad \Delta_{\text{model}}(t) := \widehat{s}_{\text{proj}}(t)^2 - \widehat{s}_{\text{spiked}}(t)^2 = \frac{\mathcal{P}_0}{\mathcal{P}(t)} - \cos^2(G_t, G_0). \tag{30}$$

Then $v(t)$ is linear in the model quantities and preserves the sign of the defect:

$$v(t) = \alpha_t\delta + \langle R_t, \tilde{G}_0\rangle_F, \qquad \Delta_{\text{model}}(t) = -\frac{2\mathcal{P}_0 v(t) + v(t)^2}{\mathcal{P}_0\,\mathcal{P}(t)}. \tag{31}$$

In particular, $v(t) = 0 \iff \Delta_{\text{model}}(t) = 0 \iff \alpha_t\delta + \langle R_t, \tilde{G}_0\rangle_F = 0$, which is a necessary scalar condition for $R_t = 0$ and $\delta = 0$ simultaneously, and when $D_{\text{chord}}(t) = 0$ ($R_t = 0$), $v(t) = \alpha_t\delta$ directly measures the orthogonality defect δ.

Proof of Theorem 4.1. For Part 1, under Assumption 2.1 ($R_t \equiv 0$), we have $\tilde{G}_t - \tilde{G}_0 = \alpha_t(\tilde{G}_T - \tilde{G}_0)$ and $\tilde{G}_{t-\Delta} - \tilde{G}_0 = \alpha_{t-\Delta}(\tilde{G}_T - \tilde{G}_0)$ with $\alpha_t, \alpha_{t-\Delta} > 0$. Since positive scalar multiples have Frobenius cosine similarity 1, $D_{\text{chord}}(t) = 1 - 1 = 0$. By strict Cauchy-Schwarz equality on $\mathcal{S}^{d\times d}$, $D_{\text{chord}}(t) = 0$ if and only if $\tilde{G}_t - \tilde{G}_0$ and $\tilde{G}_{t-\Delta} - \tilde{G}_0$ are positively collinear.

For Part 2, taking the Frobenius inner product of $\tilde{G}_t - \tilde{G}_0 = \alpha_t(\tilde{G}_T - \tilde{G}_0) + R_t$ with $\tilde{G}_0$ yields $v(t) = \langle \tilde{G}_t - \tilde{G}_0, \tilde{G}_0\rangle_F = \alpha_t\delta + \langle R_t, \tilde{G}_0\rangle_F$. Substituting $\langle \tilde{G}_t, \tilde{G}_0\rangle_F = \mathcal{P}_0 + v(t)$ into $\cos^2(G_t, G_0) = \frac{\langle \tilde{G}_t, \tilde{G}_0\rangle_F^2}{\mathcal{P}_0\mathcal{P}(t)}$ gives:

$$\cos^2(G_t, G_0) = \frac{(\mathcal{P}_0 + v(t))^2}{\mathcal{P}_0\,\mathcal{P}(t)} = \frac{\mathcal{P}_0}{\mathcal{P}(t)} + \frac{2\mathcal{P}_0 v(t) + v(t)^2}{\mathcal{P}_0\,\mathcal{P}(t)},$$

which rearranges to Equation (31). Since $\langle \tilde{G}_t, \tilde{G}_0\rangle_F = \mathcal{P}_0 + v(t) \geq 0$, the quadratic $2\mathcal{P}_0 v(t) + v(t)^2 = v(t)(2\mathcal{P}_0 + v(t))$ vanishes on $v(t) \geq -\mathcal{P}_0$ if and only if $v(t) = 0$. □

5 Window-Averaged Asymptote Extrapolation and Self-Calibrated Stopping

Intuition for Single-Exponential Linearization, Regime-Free Estimation, and Diagnostics. To overcome Corollary 3.6(3)—that $\widehat{s}_{\text{spiked}}(t)$ asymptotes to $\sqrt{1 - r/d_{\text{eff}}} < 1$ depending on the unknown terminal rank r—we model the weight-decay driven relaxation of α_t on $[t_0, \infty)$ as $\alpha_t = 1 - ce^{-\kappa(t-t_0)}$, where $c := 1 - \alpha_0 \in (0, 1]$ accounts for any initial offset $\alpha_{t_0} = \alpha_0$ accumulated during the anchor window $[t_{\text{mem}}, t_0]$. Under this relaxation:

- When $\delta = 0$, $\mathcal{P}(t) - \mathcal{P}_0 = \alpha_t^2(\mathcal{P}_T - \mathcal{P}_0)$ is a sum of two exponentials ($e^{-\kappa(t-t_0)}$ and $e^{-2\kappa(t-t_0)}$), but its square root $u(t) := \sqrt{\mathcal{P}(t) - \mathcal{P}_0} = \alpha_t\sqrt{\mathcal{P}_T - \mathcal{P}_0}$ is an *exact single exponential* $u_\infty(1 - ce^{-\kappa(t-t_0)})$.
- When $\delta \neq 0$, $u(t)$ is no longer a single exponential (and is undefined inside the $\delta < 0$ dead zone). However, the linear drift observable $v(t) = \langle \tilde{G}_t, \tilde{G}_0\rangle_F - \mathcal{P}_0 = \alpha_t\delta$ from Theorem 4.1(2) is an *exact single exponential* $v(t) = \delta(1 - ce^{-\kappa(t-t_0)})$ for all $t \geq t_0$.
- **Unified Regime-Free Route to $\widehat{\alpha}_t$:** When $w \ll 1/\kappa$ so that $\alpha_0 = 0$ ($c = 1$), $\alpha_t = 1 - e^{-\kappa(t-t_0)}$ is determined solely by κ. Once $\widehat{\kappa}(t)$ is estimated, setting $\widehat{\alpha}_t = 1 - e^{-\widehat{\kappa}(t)(t-t_0)}$ yields $\widehat{\delta}(t) = v(t)/\widehat{\alpha}_t$ and $\widehat{u}_\infty(t) = u(t)/\widehat{\alpha}_t$ directly. This avoids dividing by $\widehat{\delta}(t)$ (which is ill-conditioned as $\delta \to 0$) and eliminates the need to switch between the $\delta = 0$ and $\delta \neq 0$ formulas. Conversely, the Aitken ratio $\widehat{\alpha}_t = v(t)/\widehat{\delta}(t)$ (or $u(t)/\widehat{u}_\infty(t)$) remains exact even when $\alpha_0 > 0$ ($c < 1$).
- **Online Falsification of Exponential Relaxation:** Assumption 5.1 is directly testable online because the window-ratio rate estimate $\widehat{\kappa}(t)$ must remain constant in t, must agree across $u(t)$ and $v(t)$ whenever both carry signal, and must satisfy $u(t)/\widehat{u}_\infty(t) = 1 - e^{-\widehat{\kappa}(t)(t-t_0)}$ when $\alpha_0 = 0$.

Assumption 5.1 (First-Order Exponential Mixing Relaxation)**.** For $t \geq t_0 = t_{\text{mem}} + w$, weight decay on the post-interpolation manifold drives the mixing coefficient α_t toward 1 at a constant relaxation rate $\kappa > 0$:

$$\alpha_t = 1 - c\,e^{-\kappa(t-t_0)}, \qquad c := 1 - \alpha_0 \in (0, 1], \tag{32}$$

where $\alpha_0 = \alpha_{t_0} \in [0, 1)$ is the anchor offset at t_0 (with $\alpha_0 = 0, c = 1$ whenever $w \ll 1/\kappa$).

Lemma 5.1 (Exact Window-Averaged Aitken Extrapolation of Single-Exponential Signals)**.** *Let $z(t) = z_\infty(1 - c\,e^{-\kappa(t-t_0)})$ for $t \geq t_0$ with non-zero asymptote $z_\infty \neq 0$, offset factor $c \in (0, 1]$,*

and rate $\kappa > 0$. Fix an averaging window width $w_{ext} > 0$ and lag $\Delta > 0$, and at any time $t \geq t_0 + w_{ext} + 2\Delta$ define the three causal window means:

$$\bar{z}_k(t) := \frac{1}{w_{ext}} \int_{t_k - w_{ext}}^{t_k} z(\tau)\, d\tau, \qquad t_k := t - (3-k)\Delta \quad (k \in \{1,2,3\}). \tag{33}$$

Then for any offset factor $c \in (0,1]$, the window residuals $z_\infty - \bar{z}_k(t)$ form an exact geometric progression with ratio $q := e^{-\kappa\Delta} \in (0,1)$, and z_∞ and κ are recovered identically by:

$$\widehat{z}_\infty(t) := \frac{\bar{z}_1(t)\bar{z}_3(t) - \bar{z}_2(t)^2}{\bar{z}_1(t) + \bar{z}_3(t) - 2\bar{z}_2(t)} = z_\infty, \qquad \widehat{q}(t) := \frac{\bar{z}_3(t) - \bar{z}_2(t)}{\bar{z}_2(t) - \bar{z}_1(t)} = e^{-\kappa\Delta}, \qquad \widehat{\kappa}(t) := -\frac{\ln \widehat{q}(t)}{\Delta} = \kappa. \tag{34}$$

Proof of Lemma 5.1. Integrating the residual $z_\infty - z(\tau) = c\, z_\infty e^{-\kappa(\tau - t_0)}$ over $[t_k - w_{\text{ext}}, t_k]$ gives:

$$z_\infty - \bar{z}_k(t) = \frac{c\, z_\infty}{w_{\text{ext}}} \int_{t_k - w_{\text{ext}}}^{t_k} e^{-\kappa(\tau - t_0)}\, d\tau = z_\infty \underbrace{\left(c\, \frac{e^{\kappa w_{\text{ext}}} - 1}{\kappa w_{\text{ext}}} \right)}_{=: C_w > 0} e^{-\kappa(t_k - t_0)}.$$

Because $t_3 - t_2 = t_2 - t_1 = \Delta$ and C_w is independent of k, the unknown offset factor c cancels out of the consecutive ratios: $z_\infty - \bar{z}_2 = q(z_\infty - \bar{z}_1)$ and $z_\infty - \bar{z}_3 = q(z_\infty - \bar{z}_2)$ with $q = e^{-\kappa\Delta} \in (0,1)$. Thus $\bar{z}_3 - \bar{z}_2 = q(\bar{z}_2 - \bar{z}_1)$ and the denominator is:

$$\bar{z}_1 + \bar{z}_3 - 2\bar{z}_2 = -(1-q)^2 (z_\infty - \bar{z}_1) \neq 0. \tag{35}$$

Equating $\frac{z_\infty - \bar{z}_3}{z_\infty - \bar{z}_2} = \frac{z_\infty - \bar{z}_2}{z_\infty - \bar{z}_1} = q$ gives $(z_\infty - \bar{z}_1)(z_\infty - \bar{z}_3) = (z_\infty - \bar{z}_2)^2$, which expands to $z_\infty^2 - (\bar{z}_1 + \bar{z}_3) z_\infty + \bar{z}_1 \bar{z}_3 = z_\infty^2 - 2\bar{z}_2 z_\infty + \bar{z}_2^2$ and solves uniquely to Equation (34). □

Theorem 5.2 (Online Identification of r, δ, Alignment $s(t)$, Alignment-Threshold Time t^*, and Relaxation Diagnostics)**.** *Under Assumptions 2.1–5.1 ($R_t \equiv 0$), let $t \geq t_0 + w_{ext} + 2\Delta$, and impose a minimum progress threshold $\widehat{\alpha}_t \geq \alpha_{\min} > 0$ (equivalently $t - t_0 \geq \tau_{\min} := \frac{1}{\widehat{\kappa}(t)} \ln(\frac{c}{1-\alpha_{\min}})$) so that division by $\widehat{\alpha}_t$ and $\widehat{\alpha}_t^2$ is well-conditioned away from t_0.*

1. ***(Orthogonal Regime $\delta = 0$ via Rooted Purity Gain $u(t)$):*** *When $\delta = 0$, the **rooted purity gain** $u(t) := \sqrt{\mathcal{P}(t) - \mathcal{P}_0}$ satisfies $u(t) = u_\infty(1 - c\, e^{-\kappa(t-t_0)})$ with $u_\infty = \sqrt{\mathcal{P}_T - \mathcal{P}_0}$. Applying Lemma 5.1 to the window means $\bar{u}_1, \bar{u}_2, \bar{u}_3$ of $u(t)$ recovers $\widehat{u}_\infty(t) = u_\infty$, $\widehat{\kappa}_u(t) = \kappa$, and:*

$$\widehat{\mathcal{P}}_T(t) := \mathcal{P}_0 + \widehat{u}_\infty(t)^2 = \mathcal{P}_T, \qquad \widehat{r}(t) := \frac{1}{\widehat{\mathcal{P}}_T(t)} = r, \qquad \widehat{s}_0(t) := \sqrt{\frac{\mathcal{P}_0}{\widehat{\mathcal{P}}_T(t)}} = s_0, \qquad \widehat{\alpha}_t := \frac{u(t)}{\widehat{u}_\infty(t)} = \alpha_t. \tag{36}$$

   *Substituting $\widehat{s}_0(t)$ into Theorem 3.4(2) gives the **Self-Calibrated Online Alignment Estimator**:*

$$\widehat{s}_{cal}(t) := \widehat{s}_{spiked}(t)\sqrt{1 - \widehat{s}_0(t)^2} + \widehat{s}_0(t)\sqrt{1 - \widehat{s}_{spiked}(t)^2} = s(t), \qquad \lim_{t \to \infty} \widehat{s}_{cal}(t) = 1. \tag{37}$$

2. ***(General Regime $\delta \in \mathbb{R}$ and Regime-Free Estimation):*** *When $\delta \neq 0$ (including inside the $\delta < 0$ purity dead zone of Lemma 3.1(2)), $v(t) = \langle \tilde{G}_t, \tilde{G}_0 \rangle_F - \mathcal{P}_0 = \delta(1 - c\, e^{-\kappa(t-t_0)})$ has asymptote $v_\infty = \delta$.*

   - Offset-Tolerant Aitken Route ($\delta \neq 0$, arbitrary $c \in (0,1]$): *Applying Lemma 5.1 to the window means $\bar{v}_1, \bar{v}_2, \bar{v}_3$ of $v(t)$ recovers $\widehat{\delta}(t) := \widehat{v}_\infty(t) = \delta$, $\widehat{\kappa}_v(t) = \kappa$, and $\widehat{\alpha}_t := v(t)/\widehat{\delta}(t) = \alpha_t$.*

- Regime-Free Rate Route (valid for all $\delta \in \mathbb{R}$ when $\alpha_0 = 0$, $c = 1$): *Using $\widehat{\kappa}(t)$, define $\widehat{\alpha}_t := 1 - e^{-\widehat{\kappa}(t)(t-t_0)} = \alpha_t$ and $\widehat{\delta}(t) := v(t)/\widehat{\alpha}_t = \delta$ (which remains smooth and well-conditioned as $\delta \to 0$ without dividing by $\widehat{\delta}(t)$).*

Under either route, for $\widehat{\alpha}_t \geq \alpha_{\min} > 0$, the displacement norm and terminal purity are recovered via:

$$\widehat{\Delta\mathcal{P}}_\delta(t) := \frac{\mathcal{P}(t) - \mathcal{P}_0 - 2v(t)}{\widehat{\alpha}_t^2} = \Delta\mathcal{P}_\delta, \qquad \widehat{\mathcal{P}}_T(t) := \mathcal{P}_0 + 2\widehat{\delta}(t) + \widehat{\Delta\mathcal{P}}_\delta(t) = \mathcal{P}_T. \tag{38}$$

Substituting these into Lemma 3.2(2) recovers the exact alignment $s(t)$ online for arbitrary $\delta \in \mathbb{R}$:

$$\widehat{s}_{\mathit{cal},\delta}(t) := \frac{\mathcal{P}_0 + \widehat{\delta}(t) + \widehat{\alpha}_t\big(\widehat{\mathcal{P}}_T(t) - \mathcal{P}_0 - \widehat{\delta}(t)\big)}{\sqrt{\mathcal{P}(t)\,\widehat{\mathcal{P}}_T(t)}} = s(t), \qquad \lim_{t\to\infty} \widehat{s}_{\mathit{cal},\delta}(t) = 1. \tag{39}$$

3. ***(Single-Scalar Reduction and Predicted Alignment-Threshold Time t^*):*** *Under Assumptions 2.1–5.1, $s(t)$ is a strictly increasing deterministic function $S(\alpha_t)$ of $\alpha_t \in [\alpha_0, 1)$, so the stopping rule $\widehat{s}_{\mathit{cal},\delta}(t) \geq 1 - \epsilon$ ($\epsilon \in (0, 1 - s(t_0))$) is equivalent to $\widehat{\alpha}_t \geq \alpha^*(\epsilon)$, where $\alpha^*(\epsilon) \in (\alpha_0, 1)$ is the unique root of $S(\alpha^*) = 1 - \epsilon$. For $\delta = 0$, $\alpha^*(\epsilon)$ is given in closed form by:*

$$\alpha^*(\epsilon, r, d_{\mathit{eff}}) = \frac{\tan\psi^*(\epsilon)}{\sqrt{d_{\mathit{eff}}/r - 1}}, \qquad \psi^*(\epsilon) := \arcsin(1-\epsilon) - \arcsin\left(\sqrt{\frac{r}{d_{\mathit{eff}}}}\right), \tag{40}$$

*while for $\delta \neq 0$, because $S(\alpha) \geq 0$ is strictly increasing on $[\alpha_0, 1)$, $\alpha^*(\epsilon)$ is the unique root in $[\alpha_0, 1)$ of the quadratic equation $[\mathcal{P}_0 + \delta + \alpha(\mathcal{P}_T - \mathcal{P}_0 - \delta)]^2 = (1-\epsilon)^2\mathcal{P}_T(\mathcal{P}_0 + 2\alpha\delta + \alpha^2\Delta\mathcal{P}_\delta)$. The **predicted alignment-threshold time** t^* at which $s(t^*) = 1 - \epsilon$ is estimated causally at time t by:*

$$\widehat{t}^*(t) := t_0 + \frac{1}{\widehat{\kappa}(t)} \ln\left(\frac{\widehat{c}(t)}{1 - \alpha^*(\epsilon)}\right), \qquad \widehat{c}(t) := (1 - \widehat{\alpha}_t)e^{\widehat{\kappa}(t)(t-t_0)}, \tag{41}$$

where $\widehat{c}(t) = 1$ when $\alpha_0 = 0$. (Note that t^* is the time at which AGOP alignment reaches $1-\epsilon$; that this alignment threshold coincides with the surge in test accuracy is the empirical premise motivating AGOP alignment tracking rather than a property proved here.)

4. ***(Online Diagnostics for Assumption 5.1):*** *Under Assumption 5.1, $\widehat{\kappa}(t)$ is invariant in time. Deviations from single-exponential relaxation are diagnosed online by the **Rate-Stationarity Defect** $D_{\mathit{exp}}(t)$, the **Cross-Signal Rate Gap** $D_{u,v}(t)$ (when both u and v carry signal), and the **Trajectory Consistency Gap** $D_\alpha(t)$ (when $\alpha_0 = 0$):*

$$D_{\mathit{exp}}(t) := \frac{|\widehat{\kappa}(t) - \widehat{\kappa}(t-\Delta)|}{\widehat{\kappa}(t)}, \quad D_{u,v}(t) := \frac{|\widehat{\kappa}_u(t) - \widehat{\kappa}_v(t)|}{\widehat{\kappa}_v(t)}, \quad D_\alpha(t) := \left|\frac{u(t)}{\widehat{u}_\infty(t)} - \left(1 - e^{-\widehat{\kappa}(t)(t-t_0)}\right)\right|, \tag{42}$$

all three of which vanish identically under Assumptions 2.1–5.1 (with $D_\alpha(t) = 0$ testing both Assumption 5.1 and $\alpha_0 = 0$).

Proof of Theorem 5.2. For Part 1, when $\delta = 0$, Lemma 3.1 gives $u(t) = \sqrt{\mathcal{P}(t) - \mathcal{P}_0} = \alpha_t\sqrt{\mathcal{P}_T - \mathcal{P}_0} = u_\infty(1 - c\,e^{-\kappa(t-t_0)})$, so Lemma 5.1 applies directly to $z(t) = u(t)$.

For Part 2, when $R_t \equiv 0$, Theorem 4.1(2) gives $v(t) = \alpha_t\delta = \delta(1 - c\,e^{-\kappa(t-t_0)})$. When $\delta \neq 0$, applying Lemma 5.1 to $z(t) = v(t)$ recovers $\widehat{\delta}(t) = \delta$, $\widehat{\kappa}_v(t) = \kappa$, and $\widehat{\alpha}_t = v(t)/\widehat{\delta}(t) = \alpha_t$ for any $c \in (0, 1]$. When $c = 1$ ($\alpha_0 = 0$), $\alpha_t = 1 - e^{-\kappa(t-t_0)}$ holds for all $\delta \in \mathbb{R}$, so $\widehat{\delta}(t) = v(t)/\widehat{\alpha}_t = \delta$

without dividing by $\widehat{\delta}(t)$. By Equation (13) of Lemma 3.1, $\mathcal{P}(t) - \mathcal{P}_0 - 2v(t) = \alpha_t^2 \Delta \mathcal{P}_\delta$, which yields Equation (38) for $\widehat{\alpha}_t \geq \alpha_{\text{min}} > 0$, and Lemma 3.2(2) yields Equation (39).

For Part 3, along the chord $\tilde{G}(\alpha) = (1-\alpha)\tilde{G}_0 + \alpha \tilde{G}_T$, the cosine similarity $S(\alpha) := \cos(\tilde{G}(\alpha), \tilde{G}_T)$ is strictly increasing on $\alpha \in [0,1]$ because $\tilde{G}_0$ and $\tilde{G}_T$ are linearly independent. For $\delta = 0$, $S(\alpha_t) = \sin(\psi(t) + \theta_0) \geq 1 - \epsilon \iff \tan\psi(t) = \alpha_t \sqrt{d_{\text{eff}}/r - 1} \geq \tan\psi^*(\epsilon)$, giving Equation (40). Solving $1 - c\, e^{-\kappa(t^* - t_0)} = \alpha^*(\epsilon)$ with $c = (1-\alpha_t)e^{\kappa(t-t_0)}$ gives Equation (41). Part 4 follows immediately from the time-invariance of κ in Lemma 5.1. □

Remark 5.1 (Conditioning of Three-Window Extrapolation on Slow Plateaus). By Equation (35), the Aitken denominator equals $-(1-q)^2(z_\infty - \bar{z}_1)$ where $q = e^{-\kappa\Delta}$. When $\kappa\Delta \ll 1$ on a slow grokking plateau, $(1-q)^2 \approx (\kappa\Delta)^2 \ll 1$, which amplifies residual variance by $\mathcal{O}((1-q)^{-2})$. In practice, to ensure well-conditioned online inversion: (i) choose the window lag $\Delta \gtrsim 1/\widehat{\kappa}$ once an initial rate estimate is available (so $q \leq e^{-1}$ and $(1-q)^{-2} \leq 2.5$), or (ii) replace the three-point algebraic solve with a causal nonlinear least-squares fit of $z_\infty(1 - e^{-\kappa(\tau - t_0)})$ across the entire observed history $\tau \in [t_0, t]$.

6 Summary of Online Observables and Theoretical Guarantees

Observable / Estimator	**Closed-Form Expression**	**Theoretical Role**
$\mathcal{P}(t)$ (Spectral Purity)	$\frac{\|G_t\|_F^2}{\text{Tr}(G_t)^2} = \frac{1}{r_{\text{eff}}(t)}$	Inverse participation ratio on $\Delta^{d\times d}$ ($\mathcal{P}_0 := \|\tilde{G}_0\|_F^2$)
$\widehat{s}_{\text{spiked}}(t)$ (Purity Estimator)	$\sqrt{1 - \frac{\mathcal{P}_0}{\mathcal{P}(t)}} = \sqrt{1 - \frac{r_{\text{eff}}(t)}{r_{\text{eff},0}}}$	Alignment with $\hat{G}_T^\perp$ under A2.1–A2.3 ($\delta = 0$); undefined in $\delta < 0$ dead zone $\mathcal{P}(t) < \mathcal{P}_0$
$\widehat{s}_{\text{proj}}(t)$ (Projection Estimator)	$\sqrt{1 - \cos^2(G_t, G_0)}$	Alignment with $\hat{G}_T^\perp$ under A2.1 for any $\delta \in \mathbb{R}$
$D_{\text{chord}}(t)$ (Chord Defect)	$1 - \cos\left(\tilde{G}_t - \tilde{G}_0, \tilde{G}_{t-\Delta} - \tilde{G}_0\right)$	Direct online falsification test of A2.1 ($D_{\text{chord}} = 0$ on chord; evaluate on window means for $\widehat{\alpha}_t \geq \alpha_{\text{min}}$)
$v(t)$ (Linear Drift)	$\langle \tilde{G}_t, \tilde{G}_0 \rangle_F - \mathcal{P}_0 = \alpha_t \delta + \langle R_t, \tilde{G}_0 \rangle_F$	Sign-preserving linear measure of orthogonality defect δ ($R_t \perp \tilde{G}_T - \tilde{G}_0$)
$D_{\text{exp}}(t), D_{u,v}(t), D_\alpha(t)$	Eq. (42)	Online falsification tests of exponential relaxation A5.1 (rate stationarity, cross-signal agreement, and offset check)
$\widehat{s}_{\text{cal}}(t), \widehat{s}_{\text{cal},\delta}(t)$ (Calibrated)	Eqs. (37) ($\delta = 0$) and (39) ($\delta \in \mathbb{R}$)	Exact online recovery of $s(t)$ and predicted alignment-threshold time $\widehat{t}^*$ under A2.1–A5.1 (for $\widehat{\alpha}_t \geq \alpha_{\text{min}}$)

Table 1: Summary of causal AGOP observables, model diagnostics, and self-calibrated estimators.